

Bio-inspired Event-frame Dual-modal Tactile Sensor for Texture and Stiffness Classification

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Abstract—Inspired by the human tactile system, which relies on fast-adapting mechanoreceptors for dynamic stimuli and slow-adapting mechanoreceptors for static sustained pressure, we propose a novel event-frame dual-modal tactile sensor. Based on the DAVIS346 camera, it synchronously captures microsecond-resolution dynamic event streams and grayscale frame videos with precise spatio-temporal alignment. We develop a three-stage fusion method encompassing frame-synchronized event accumulation, independent single-modal training via a shared Transformer architecture, and dynamic decision-level fusion. We evaluate the sensor and the network on two highly challenging tactile tasks. In a texture discrimination task involving lateral sliding on ten fabrics with different compositions or weaving structures, the dual-modal approach achieves 91.67% accuracy, outperforming the best single modality by 13.34%. In a stiffness classification task involving vertical pressing on ten polyurethane foams, the dual-modal approach achieves 98.92%, outperforming the best single modality by 6.47%. The results demonstrate that the combination of event and frame modalities significantly enhances the capability of robotic tactile systems on distinguishing subtle material differences.

I. INTRODUCTION

Tactile sensing is critical for robotic precision manipulation, object recognition, and human-robot interaction. In recent years, robotic tactile sensing has garnered significant attention and rapid development across various domains including industrial automation, service robotics, and medical robotics [1]. However, existing robotic tactile perception remains considerably limited compared to the human tactile system, particularly in perceiving complex textures and subtle deformations.

The human tactile system achieves remarkable perception through a synergistic dual-pathway mechanism [2], combining fast-adapting mechanoreceptors for dynamic stimuli (e.g., textures and vibrations) and slow-adapting ones for static pressure. Existing tactile sensors typically rely on single sensing principles. Although combining elements can emulate fast-slow responses [3], they still cannot fully replicate the sophisticated human fast-slow mechanism, thereby

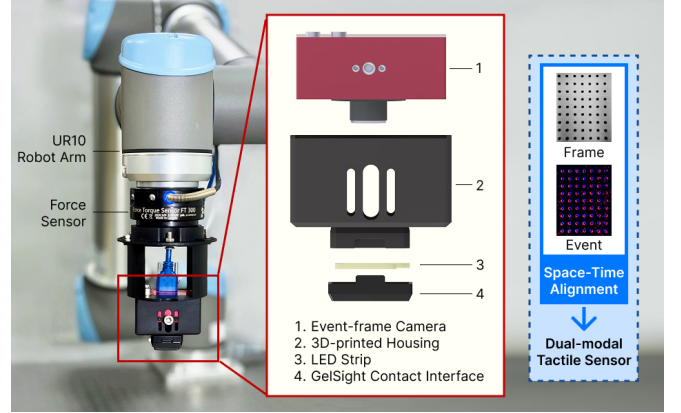


Fig. 1. Our novel event-frame dual-modal tactile sensor, achieving synchronized acquisition of spatio-temporally aligned frame images and event streams (red and blue for positive and negative events). This hardware-level implementation authentically replicates the fast-slow dual-pathway perception mechanism of the human tactile system.

failing to support complex and dexterous manipulation tasks like humans.

As bio-inspired sensors, event cameras asynchronously record pixel brightness changes with microsecond resolution and high dynamic range, perfectly aligning with the high-frequency response of human fast tactile perception. Meanwhile, conventional frame cameras provide rich static image information, echoing human slow tactile perception for stable contact. Based on this innovative cross-modal analogy, as shown in Fig. 1, we present a novel event-frame dual-modal robotic tactile sensor, which, to the best of our knowledge, is the first to replicate the fast-slow dual-pathway perception mechanism of human tactile system by the combination of event and frame modality. The main contributions of this work include:

- Development of a novel event-frame dual-modal tactile sensor, capturing dynamic and static tactile information to achieve hardware-level emulation of human fast-slow tactile mechanisms.
- Proposal of a novel dual-modal fusion network for tactile perception that fully leverages the complementary advantages of event and frame modalities, significantly enhancing tactile perception accuracy.
- Validation of the proposed sensor and network through fabric texture and foam stiffness discrimination experiments. By employing ten materials with subtle physical differences, the dual-modal method achieved 98.92% and 91.67% accuracy, outperforming the best single modality by 6.47% and 13.34%, respectively.

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II. RELATED WORK

The core capabilities of tactile perception can be characterized along two critical dimensions: stiffness perception and texture perception. As a critical modality for human environmental perception, tactile information possesses unique advantages that vision cannot provide. While vision captures rich appearance information, it cannot directly perceive physical properties such as object compliance and surface roughness. Psychophysical studies [4] have demonstrated that stiffness and texture roughness constitute the two most important independent dimensions in tactile perception, thus representing the core task and key challenge of robotic tactile sensors.

Robotic tactile sensors are generally categorized into non-vision-based and vision-based types. Traditional non-vision-based sensors, such as piezoelectric [5], piezoresistive [6], and capacitive sensors [7] perceive object properties by measuring contact forces, pressure, or vibrations. While some of these sensors can address both stiffness and texture tasks, they often struggle to move beyond classifying distinct material categories to identifying subtle intra-class variations. For instance, the capacitive sensor by Qiu et al. [8] distinguishes various fabric textures, yet its stiffness recognition remains limited to cases with significant differences, such as wood versus sponge. Although BarrettHand [9] demonstrates excellent stiffness discrimination on a challenging dataset of ten polyurethane foams [10], its experiments regarding texture recognition are not involved.

However, in the realm of vision-based sensors, there may exist a pathway to identify subtle intra-class variations on both stiffness and texture tasks. Vision-based tactile sensors record contact images via cameras, include frame-based sensors [11] and event-based sensors [12]. For texture tasks, Taunyazov et al. [12] designed an event-based system to distinguish frosted bottles and a metal cans. Both of the methods proposed by Belousov et al. [13] and Ward-Cherrier et al. [14] can successfully classify fabrics of varying textures. For stiffness tasks, GelSight can distinguish silicone blocks of different compositions [15], while Han et al. [16] use a series of visuotactile sensors to construct stiffness distribution maps of objects. Table I summarizes the performance of representative vision-based tactile sensors on stiffness and texture discrimination tasks. “✓” indicates that the work involves relevant experiments, while “×” indicates it does not.

As shown in Table I, existing single-modal sensors typically excel at either stiffness or texture discrimination, but not both. In contrast, our dual-modal sensor fuses comple-

mentary advantages of both modalities to achieve superior performance in both dimensions, validating the effectiveness of the dual-modal perception mechanism.

III. METHODOLOGY

A. Event-frame Dual-modal Tactile Sensor Design

The proposed event-frame dual-modal tactile sensor, as illustrated in Fig. 1, comprises four core components: a DAVIS346 event-frame dual-modal camera, a 3D-printed housing, an LED strip, and a GelSight [17] contact interface. The core component is DAVIS346 camera, which simultaneously outputs event streams and frame-based video at a resolution of 346×260 pixels, with the frame modality operating at 40 frames per second (FPS) and the event modality achieving 1 microsecond temporal resolution. This camera enables precise spatio-temporal alignment between the two modalities. Spatially, the event and frame imaging regions are perfectly co-located on the same sensor array without any viewpoint discrepancy, ensuring pixel-level correspondence. Temporally, while the frame modality continuously captures images and the event stream only outputs when brightness changes occur, both modalities respond synchronously to the same physical events.

We utilize a GelSight contact interface embedded with a dot-patterned elastomer. When the sensor contacts an object, the dot pattern deforms and is captured by the camera, providing rich tactile information. To enable clear and stable imaging of the contact interface, we designed and 3D-printed a black resin housing with an integrated LED strip for uniform illumination. The housing is equipped with a snap-fit mechanism at the front end to securely hold the GelSight contact interface. For a more compact structure, we use a small 4.2 mm fixed-focus lens, and adjust its position through a slot on the housing to achieve proper focus. The dual-modal tactile sensor is integrated with a UR10 robotic manipulator equipped with a force-torque sensor.

B. Event-frame Dual-modal Fusion Method

As illustrated in Fig. 2, our event-frame dual-modal fusion method consists of three sequential stages: data preprocessing, single-modal training, and dual-modal ensemble fusion.

Data Preprocessing. We exploit the hardware-level event-frame synchronization of the DAVIS346 camera to generate spatio-temporally aligned training samples. As Fig. 2(a) shows, For every two consecutive frames with timestamps t_i and t_{i+1} , we extract all events occurring within $[t_i, t_{i+1}]$ and uniformly accumulate them into four event frames. This frame-timestamp-based accumulation preserves dual-modal synchronization and achieves pixel-level and temporal-level alignment. Accumulating into four frames rather than one avoids polarity cancellation while preserving temporal dynamics.

Data sampling includes both force-peak-centered sliding window sampling and direct sampling. The former centers the sliding window around force peaks, while the latter samples directly from the continuous data stream. For certain

TABLE I

COMPARISON OF VISION-BASED TACTILE SENSORS.

Reference	Modality	Stiffness	Texture
[12]	Event	×	✓
[15]	Frame	✓	×
[13]	Frame	×	✓
[14]	Event	×	✓
[16]	Frame	✓	×
OURS	Event+Frame	✓	✓

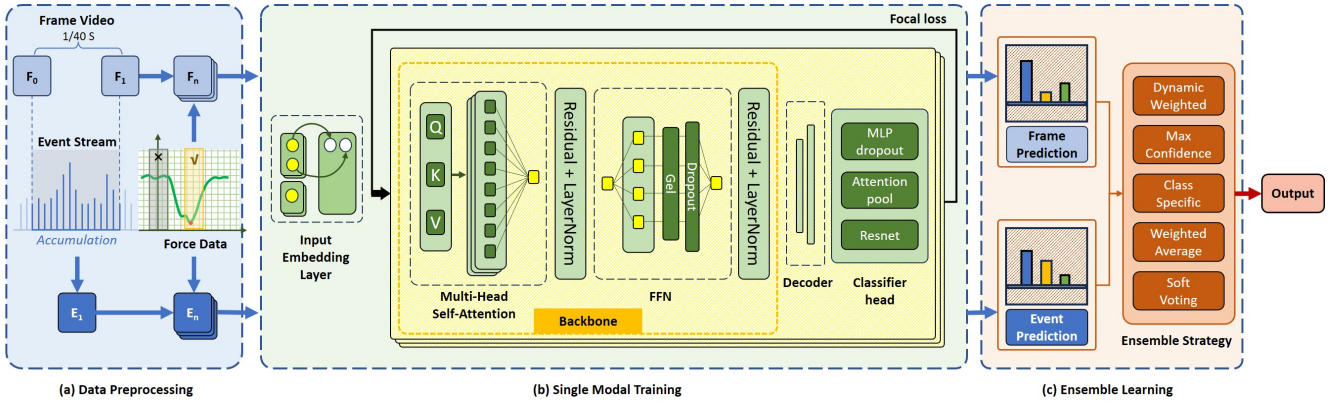


Fig. 2. Overview of the event-frame dual-modal fusion method. (a) Data preprocessing stage performs frame-synchronized event accumulation on frame video and event streams, followed by force-peak-centered sliding window sampling or direct sampling. (b) Single-modal training stage. Both event and frame share the same network architecture, sequentially passing through input embedding layer, Transformer backbone with multi-head self-attention mechanism, FFN (feed-forward network), decoder, and multiple classification heads, with focal loss employed to address class imbalance. (c) Decision-level ensemble fusion stage combines prediction probabilities from frame and event modalities through multiple ensemble strategies, dynamically selecting the optimal strategy to derive the final prediction.

tactile tasks, the moment of contact contains the richest tactile information. By centering around force peaks and setting an appropriate window size to include both the just-contact and just-release time points, we can significantly reduce interference from irrelevant information and enhance model performance.

Single-Modal Training. Event and frame modalities are trained independently using a shared network architecture. As Fig. 2(b) shows, single-modal training consists of four modules: Input Embedding Layer, Backbone, Decoder, and Classifier Head. The Input Embedding Layer divides input images into patches and performs linear embedding, while adding Fourier positional encoding to preserve spatial location information. After data augmentation with 15% random masking, the features undergo spatio-temporal modeling through a latent memory module [18] before being fed into the Transformer backbone.

Within the backbone, the multi-head self-attention mechanism captures multi-perspective features including local texture characteristics, global contact patterns, and temporal dynamics. The FFN (feed-forward network) [19] further enhances feature representation capability. The network is optimized using the AdamW optimizer [20] with cosine annealing learning rate scheduling and an early stopping mechanism. After decoding, features are fed into multiple classification heads such as MLP dropout [21], Attention Pool [22], and ResNet [23]. Due to class imbalance which commonly presents in tactile data, we employ Focal Loss, allowing the network to automatically focus on hard-to-classify samples. After single-modal training, the network saves the prediction probabilities for each class for dual-modal ensemble learning.

Dual-Modal Ensemble Fusion. Decision-level fusion is employed to combine predictions from both modalities as shown in Fig. 2(c). Feature-level fusion approaches that directly merge features from both modalities during training may cause the weaker modality to introduce noise and

impair the learning of the stronger modality and potentially result in fused model performance inferior to the best single modality. Decision-level fusion trains each modality independently to largely avoid this issue. By dynamically selecting more reliable predictions based on each modality’s strengths, decision-level fusion rarely performs worse than the best single modality, effectively mitigating the risk of performance degradation during fusion.

We implement five ensemble strategies. Max Confidence selects higher-confidence modality, Weighted Average performs probability weighting on overall single-modal accuracies, Dynamic Weighted adaptively adjusts weights with prediction entropy and Soft Voting simply averages probabilities [24], [25]. Our refined Class-Specific strategy evolves from the foundational concept of class-dependent decision fusion [26] by incorporating an automated modality-preference labeling mechanism based on class-wise validation accuracies.

Our refined Class-Specific strategy first automatically analyzes class-wise modality performance, then labels a preferred modality for each class by marking it as preferred if one modality’s accuracy substantially exceeds the other or as both if they perform similarly. During inference, predictions from both modalities are directly adopted when they agree, while for disagreements weighted probability fusion is employed with weights dynamically allocated proportional to historical accuracy ratios. When accuracy differences exceed a substantial threshold, extreme weights such as 0.95 versus 0.05 are applied, and when one modality achieves perfect accuracy on its predicted class while the other performs significantly worse, that modality’s prediction is directly trusted. This mechanism enables the network to automatically learn which modality to trust for each class. During testing, all strategies are evaluated and the optimal fusion method is automatically selected.

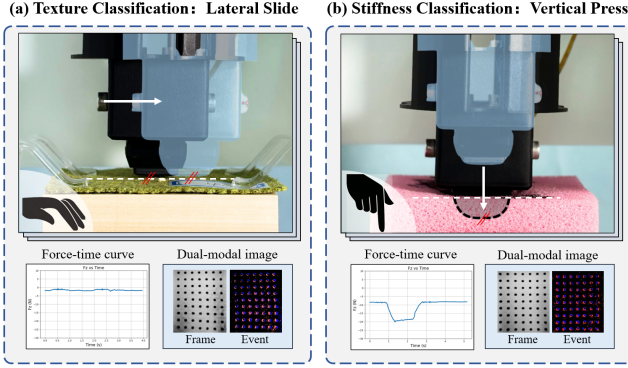


Fig. 3. Biomimetic tactile exploration design inspired by typical human manual exploration patterns for perceiving texture and stiffness. (a) Lateral Slide. The sensor slides across the sample surface to do texture tasks. (b) Vertical Press. The sensor penetrates into the sample to do stiffness tasks. Force-time curves represent the Z-axis force profiles throughout the motion, and dual-modal images show the corresponding event and frame modality responses.

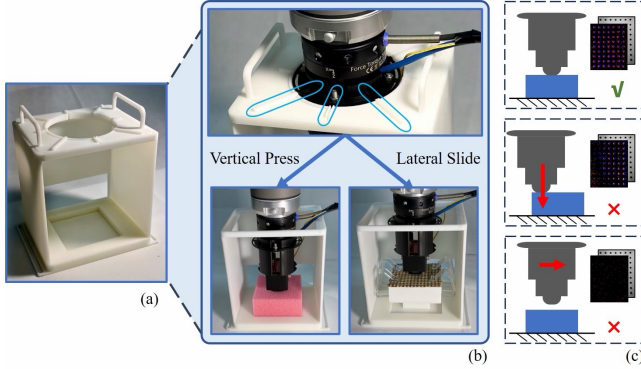


Fig. 4. (a) Physical experimental setup comprising two radially patterned plates and a support frame. (b) The sensor's tracking patterns align with the plates for determining Z-axis height in lateral sliding and centering in vertical pressing. Both plates are then removed via handles to prevent motion interference during robotic manipulation. (c) Acquisition of uniform dual-modal images without missing regions.

IV. EXPERIMENTS

A. Experiment Design

Biomimetic Exploration Strategy. During human tactile exploration, texture perception typically involves tangential sliding, whereas stiffness evaluation necessitates substantial normal indentation. To replicate this human tactile exploration, we designed a biomimetic robotic strategy as shown in Fig. 3. For texture discrimination, the manipulator translates horizontally along the X-axis to simulate lateral sliding with gentle force. Event images exhibit asymmetric optical flow displacement, while frame images show tangential deformation. For stiffness discrimination, the manipulator descends vertically along the Z-axis until reaching a peak normal force, emulating vertical pressing. Event images display radially symmetric optical flow expansion, while frame images exhibit radially symmetric deformation.

To implement the lateral sliding and vertical pressing contact modes, we designed a unified control method. For the lateral sliding mode, the manipulator translates along the X-axis at a constant velocity for a distance D before returning.



Fig. 5. Ten polyurethane foam specimens with different Shore 00 hardness values.

TABLE II
FOAM STIFFNESS SPECIFICATIONS

Label	1	2	3	4	5	6	7	8	9	10
Shore 00	44.0	19.0	38.0	28.5	36.0	50.5	40.0	53.5	33.0	49.0

For the vertical pressing mode, the manipulator descends vertically along the Z-axis while monitoring the normal force F_z . When F_z exceeds the threshold F_{th} or the distance reaches D , the manipulator pauses for a holding duration T before returning to the initial position. This prevents damage while ensuring the acquired data exhibits distinct descending, holding, and ascending phases.

Experimental Setup. Precise initial positioning is critical for experimental repeatability. In the lateral sliding mode, contact force is entirely determined by the initial Z-axis height. Insufficient contact yields no actionable tactile data, whereas excessive initial normal force can cause thin fabric specimens to roll up or wrinkle during X-axis traversal. Conversely, the vertical pressing mode requires perfectly centered alignment. Off-center indentation fails to accurately reflect intrinsic material properties and tends to cause missing regions in the captured images.

We designed a physical experimental setup as shown in Fig. 4, comprising two plates with radial patterns and a support frame. Prior to experiments, the sensor's embedded pattern is aligned with the plate patterns to ensure the contact interface is centered and aligned with the specimen while establishing the optimal Z-axis initial height. The plates are then removed before robotic manipulation. For lateral sliding, a dedicated fixture base with four clamps secures the fabrics at a meticulously engineered height to achieve the required gentle contact. For vertical pressing, specimens are constrained within a perfectly centered bottom groove. When properly aligned, the acquired dual-modal images exhibit no missing regions.

B. Stiffness Classification Experiment

The stiffness classification experiment aims to validate the performance of our event-frame dual-modal tactile sensor in distinguishing materials with different stiffness levels. We selected ten polyurethane foam specimens with varying stiffness, as shown in Fig. 5. These specimens share the same material composition but differ in density. We measured their stiffness using a Shore durometer, as detailed in Table II. The Shore 00 hardness values range from 19.0 to 53.5, representing extremely subtle differences that pose significant challenges for tactile-based discrimination.

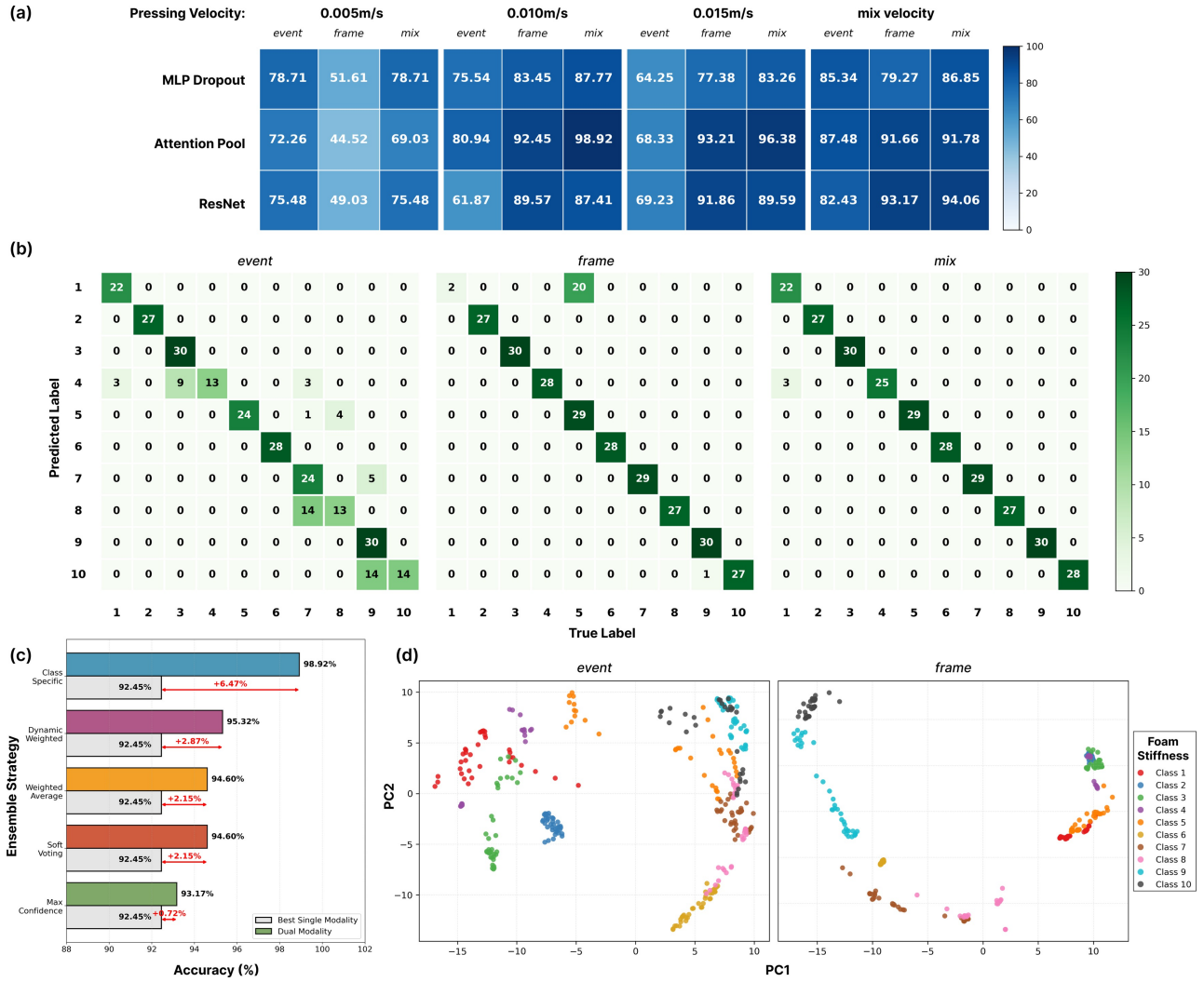


Fig. 6. Stiffness classification experiment results. (a) Classification accuracy of event, frame and mix modalities under different velocities and classifier heads. (b) Confusion matrices for each modality and mix modalities under the best combination of velocity, classifier head and ensemble strategy. The best single modality: frame. (c) Accuracy comparison of different ensemble strategies under the best combination of velocity and classifier head. (d) Principal Component Analysis on the learned features under the best combination of velocity and classifier head.

Data Collection. We conducted experiments at three different indentation velocities: low speed (0.005 m/s), medium speed (0.010 m/s), and high speed (0.015 m/s), performing 15 independent trials for each velocity condition. For data processing, we employed a force-peak-based selective sampling strategy, each trial yields approximately 5–8 valid samples. After data collection and outlier filtering, approximately 100 samples per class were retained for training and testing.

Training Details. During single-modal training, event and frame modalities are trained independently. The dataset is split into 70% training and 30% validation. Event modality uses 100 time steps and frame modality uses 50 time steps. Both modalities are trained for 40 epochs. We evaluated three different classification head architectures: MLP dropout, Attention Pool, ResNet. The optimizer is AdamW with learning rate 3×10^{-4} and weight decay 1×10^{-4} . Learning rate scheduling employs cosine annealing with initial period $T_0 = 10$, period multiplication factor $T_{mult} = 2$, and

minimum learning rate $\eta_{min} = 5 \times 10^{-5}$. Early stopping is implemented with patience of 15 epochs and minimum improvement threshold of 0.001. The loss function is Focal Loss with a focusing parameter $\gamma = 2.0$. In the dual-modal ensemble fusion stage, the network evaluates all strategies and automatically selects the one with the best performance on the test set as the final prediction.

Classification Result. As shown in Fig. 6, the experiment evaluated the classification performance of three single-velocity strategies and a mixed-velocity strategy. The highest event accuracy of 87.48% was achieved using the attention pool head under mixed-velocity training, while the highest frame accuracy reached 93.21% at 0.015 m/s using the same head. Ultimately, the model achieved the optimal bimodal accuracy of 98.92% at 0.010 m/s using the attention pool head. Under this optimal bimodal condition, the individual event and frame accuracies were 80.94% and 92.45%, respectively. Therefore, the ultimate bimodal performance is

determined not only by the training effectiveness of the two single modalities, but also by whether they exhibit strong complementarity during training.

Velocity Analysis. At a low velocity of 0.005 m/s, the event modality significantly outperformed the frame modality. This occurs because low-speed motion causes frame data to overlap heavily and lack effective features, whereas the event modality acutely captures subtle physical deformations. When the velocity increase, frame modality surpassed event modality. High-speed pressing induces high-frequency noise in the event modality, enabling the frame modality to acquire more complete information. The performance of both modalities tended to balance during mixed-velocity training. However, low-quality data samples from low-velocity conditions prevented the overall accuracy from exceeding the optimal level achieved at the medium velocity.

Classifier Head Comparison. Attention pool performed poorly due to poor-quality frame inputs at low velocities. The attention mechanism failed to correctly allocate feature weights under these conditions. Conversely, MLP dropout and ResNet demonstrated superior robustness. MLP dropout’s simple linear mappings provide stability by avoiding complex feature interactions that amplify noise. ResNet’s uniform feature processing ensures consistent extraction without being misled by unreliable weights. These simpler architectures effectively trade adaptability for robustness, making them more suitable for degraded data. At higher velocities, input information became richer and attention pool’s advantages emerged. MLP dropout risks losing crucial spatio-temporal structure in complex scenarios. ResNet struggles to adaptively capture localized core characteristics. Attention pool dynamically assigns weights based on feature importance. This mechanism enables selective focus on informative contact deformation regions while suppressing background noise. When provided with high-quality data, it naturally outperforms the other two architectures.

Dual-Modal Complementarity. Under the optimal configuration, as shown in Fig. 6(b) and (c), the event and frame modalities achieved accuracies of 80.94% and 92.45%, respectively. Bimodal fusion reached 98.92%, yielding a 6.47% improvement by Class Specific ensemble strategy. The misclassified categories of the two modalities exhibit almost no overlap. Frame modality produces high-confidence but systematically biased predictions, such as misclassifying nearly all samples of Class 1 as Class 5. This occurs because frame modality relies on static characteristics. When two materials share similar static characteristics, the frame modality consistently confuses them. Conversely, event modality generates scattered and uncertain cross-category predictions, which crucially include the correct classifications entirely missed by the frame modality. This results from event modality’s sensitivity to dynamic responses and high-frequency variations. While this sensitivity introduces uncertainty under noise, it enables discrimination based on transient behaviors. Materials sharing static appearance may exhibit distinct dynamic signatures that event modality captures.

Furthermore, as illustrated by the PCA plots in Fig. 6(d), the same material category occupies distinctly different regions within the event and frame feature spaces, indicating that the two modalities encode different physical information. Crucially, certain classes form tight clusters in event space but disperse in frame space, and vice versa. This cross-modal cluster quality variation directly explains why classes well-separated in one modality correspond to frequent misclassifications in the other. Such spatial complementarity fundamentally enables decision-level fusion to integrate advantageous predictions from both modalities, significantly outperforming single-modality configurations.

C. Texture Classification Experiment

Texture classification experiment aims to validate the performance of our event-frame dual-modal tactile sensor in distinguishing materials with different textures. As shown in Fig. 7, we selected ten fabric specimens with diverse fiber compositions and weaving structures. As detailed in Table III, these specimens encompass various fiber compositions and weaving structures.



Fig. 7. Ten fabric specimens with diverse fiber compositions and weaving structures.

TABLE III
FABRIC COMPOSITION & STRUCTURE SPECIFICATIONS

Label	Composition	Structure
1	100% Cotton	Houndstooth weave
2	100% Linen	Plain weave
3	100% Tencel	Satin weave
4	78% Nylon, 22% Spandex	Double rib knit
5	100% Polypropylene	Nonwoven
6	96% Cotton, 4% Spandex	Rib knit
7	70% Polyester, 30% Viscose	Twill weave
8	100% Polyester	Satin weave
9	65% Polyester, 35% Cotton	Plain weave
10	90% Wool, 10% Polyester	Twill weave

Data Collection. All experiments were conducted at a sliding velocity of 0.015 m/s, with three independent trials performed for each specimen. Since the sensor contacts different regions of the fabric during X-axis sliding motion, we employed a denser direct sliding-window sampling strategy for data processing, yielding over 20 valid samples per trial. Approximately 100 samples per class were retained for training and testing. Texture classification experiment maintains the same network configuration as the stiffness classification experiment.

Classification Result. As shown in Fig. 8(a) and (b), with attention pool, event and frame modalities achieved accuracies of 73.75% and 78.33%, and dual-modal approach

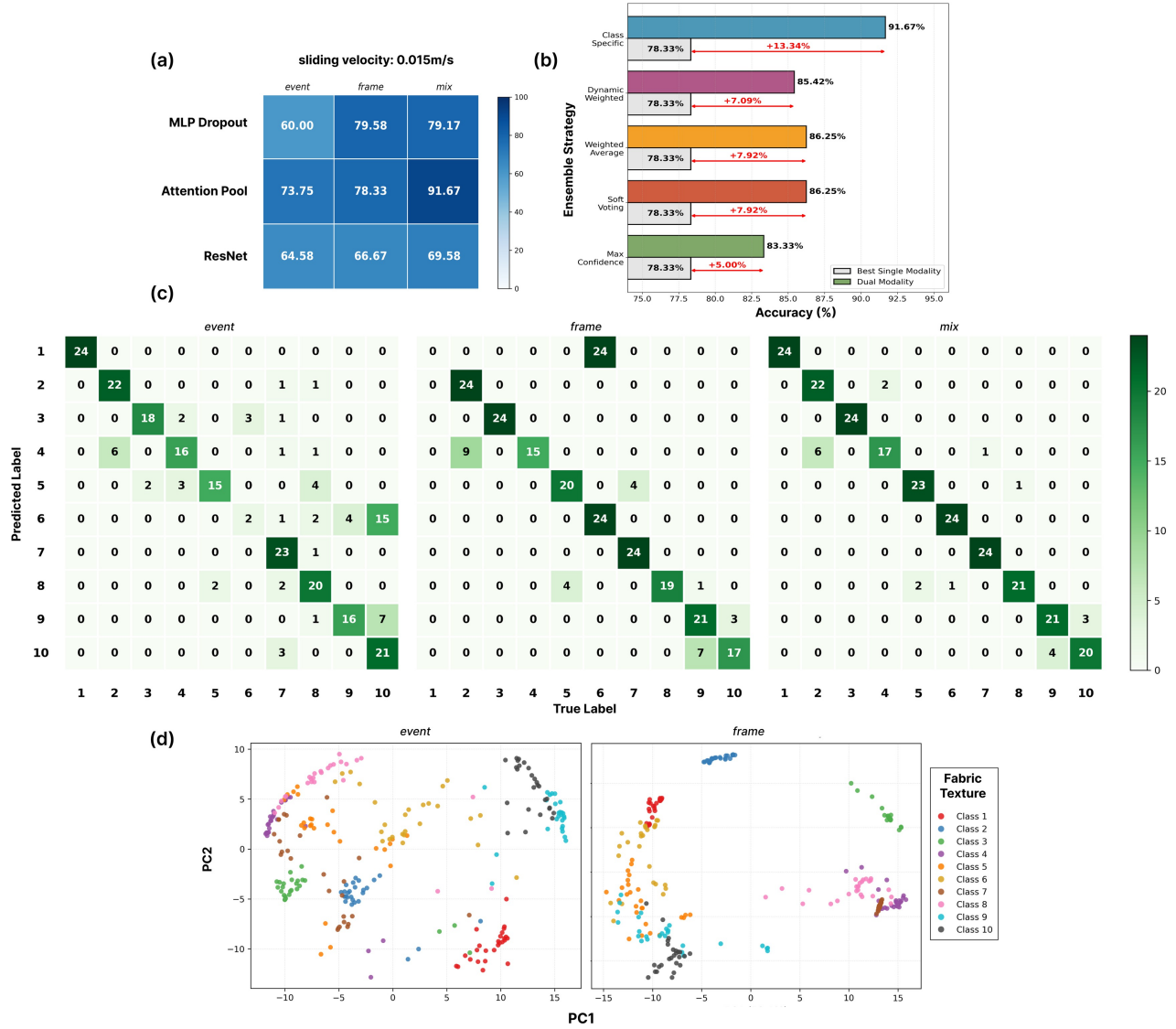


Fig. 8. Texture classification experiment results. (a) Classification accuracy of event, frame and mix modalities under different classifier heads. (b) Accuracy comparison of different ensemble strategies under the best classifier head. (c) Confusion matrices for each modality and mix modalities under the best combination of classifier head and ensemble strategy. The best single modality: frame. (d) Principal Component Analysis on the learned features under the best classifier head.

achieved the best accuracy of 91.67%. the Class Specific ensemble strategy yields a 13.34% improvement over the best single modality. For other classification heads, neither single-modal nor dual-modal accuracies exceeded 80%, indicating that the attention pool head is significantly superior to the other two architectures in this task.

Ensemble Strategy. Fig. 8(c) illustrates that the two modalities excel at different categories with attention pool, providing complementary information. This allows the Class Specific strategy’s mechanism of automatically analyzing each category’s modality performance to intelligently select the optimal modality for each class. However, with other classification heads such as resnet, the performance of the two modalities tends to be more similar, lacking clear complementarity. As a result, the advantage of dual-modal fusion cannot be fully realized, and may even be

negatively impacted by high-confidence misclassifications, leading to performance that is slightly worse than the best single modality.

PCA Topology Visualization. Fig. 8(d) and Fig. 6(d) reveal key characteristics of the dual-modal feature space. Different material classes form distinct clusters, demonstrating that the network successfully learns discriminative features. Notably, both foam and fabric experiments exhibit remarkable ring-like linear distributions. The linear alignment indicates that the samples are arranged in an orderly manner governed by specific underlying physical laws such as continuous variations in stiffness or surface roughness. The ring-like topology suggests that this is likely a projection mapping of complex representations from a high-dimensional feature space. This thoroughly confirms that the system essentially captures intrinsic physical properties

of the materials, rather than incidental imaging factors like lighting variations or sensor position deviations.

V. CONCLUSION

This study implemented an event-frame dual-modal tactile sensor based on the DAVIS346 camera, achieving hardware-level emulation of the human fast-slow tactile mechanism. Through biomimetic tactile exploration behaviors, dual-modal fusion achieved 98.92% and 91.67% accuracy on ten foam and ten fabric discrimination tasks, improving upon the best single modality by 6.47% and 13.34%, respectively. The event modality captures dynamic deformation and high-frequency vibration, while the frame modality captures static deformation distribution. Their complementarity enables decision-level fusion to comprehensively leverage multi-dimensional information for superior performance, providing a powerful technical pathway for robotic tactile perception.

In terms of limitations, while we validated the system's effectiveness on two core tactile tasks, the experimental scale remains limited, future work should validate the system on a larger material library. Besides, the current system primarily targets material recognition tasks. Future work can expand to more complex application scenarios such as tumor detection and fruit ripeness recognition, which require finer-grained tactile perception and more complex decision-making mechanisms.

VI. ACKNOWLEDGMENT

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